Package ‘Rmutaphy’

February 9, 2026

Type Package

Title Detecting Shared Mutations Associated with a Trait in a Phylogenetic Framework

Version 0.1.0

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Description Implements MutaPhy, a phylogeny-based method for detecting shared mutations associated with a trait by identifying phenotype-enriched clades in a phylogenetic tree. The method relies on permutation-based statistical tests and ancestral sequence reconstruction, and was developed and evaluated using simulated data as well as applied to dengue virus genomic and clinical datasets.

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Encoding UTF-8

LazyData true

RoxygenNote 7.3.1

Imports ape,
phangorn,
stats

Contents

`ai_test` *Permutation test for Association Index (AI)*

Description

Computes the observed Association Index (AI) and assesses significance using a permutation test.

Usage

```
ai_test(tree, trait, n_simu = 1000)
```

Arguments

`tree`	A phylogenetic tree of class `"phylo"`.
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.
`n_simu`	Number of permutations (default: 1000).

Details

AI is computed by summing node-wise contributions that depend on the majority state frequency among descendants and the number of descendant tips. Lower AI indicates stronger clustering. See Wang and al, 2001.

Value

A list with:

ai_obs Observed AI value.

pvalue Permutation p-value (numeric scalar).

null_distribution Numeric vector of permuted AI values.

Note

In `phylo.stats()`, `ai_test()` is called without passing `n_simu` (so it uses its default 1000), even if `n_simu` is set in `phylo.stats()`.

References

Wang and al (2001). Identification of Shared Populations of Human Immunodeficiency Virus Type 1 Infecting Microglia and Tissue Macrophages outside the Central Nervous System.

See Also

`calculate_AI`

Examples

```
## Not run:
set.seed(1)
tree <- ape::rcoal(30)
trait <- sample(c(rep("severe", 15), rep("non severe", 15)))
# trait must be ordered as tree$tip.label:
# here it's already length Ntip(tree) and will be interpreted in that order.

res_ai <- ai_test(tree = tree, trait = trait, n_simu = 100)
res_ai$pvalue

## End(Not run)
```

---

calculate_AI — *Compute Association Index (AI) on a tree*

Description

Computes an Association Index (AI) by iterating over internal nodes and summarizing trait clustering among descendant tips.

Usage

```
calculate_AI(tree, trait)
```

Arguments

`tree`	A phylogenetic tree of class "phylo".
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.

Details

For each internal node, the function retrieves descendant tips using `get_tips_descendants` and adds a node-wise contribution based on the maximum trait frequency among descendants and the descendant clade size. See Wang and al, 2001.

Value

Numeric scalar, the AI value.

References

Wang and al (2001). Identification of Shared Populations of Human Immunodeficiency Virus Type 1 Infecting Microglia and Tissue Macrophages outside the Central Nervous System.

See Also

`get_tips_descendants`, `ai_test`

calculate_corr_pvalues_perm
Computes corrected permutation p-values after hierarchical correction by aggregating the corrected binary permutation matrix.

Description

Computes corrected permutation p-values after hierarchical correction by aggregating the corrected binary permutation matrix.

Usage

```
calculate_corr_pvalues_perm(corrected_matrix)
```

Arguments

`corrected_matrix`	A binary matrix after hierarchical correction, as produced within `mutaphy_test()`.

Value

A data.frame with corrected permutation p-values for each node.

calculate_MC *Compute Monophyletic Clade size (MC) for two states*

Description

For each of the two states (`trait1` and `trait0`), computes the size of the largest internal clade whose descendant tips all share that state (i.e., a monophyletic clade for that state).

Usage

```
calculate_MC(tree, trait, trait0, trait1)
```

Arguments

`tree`	A phylogenetic tree of class `"phylo"`.
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.
`trait0`	Character scalar: label for state "0".
`trait1`	Character scalar: label for state "1".

Details

If a state is absent from `trait`, its MC is returned as 0. See Salemi and al, 2005.

Value

Named numeric vector of length 2 with MC values for `trait1` and `trait0`.

References

Salemi and al (2005). Phylodynamic analysis of human immunodeficiency virus type 1 in distinct brain compartments provides a model for the neuropathogenesis of AIDS.

See Also

get_tips_descendants, mc_test

---

calculate_PS *Compute Parsimony Score (PS) for a binary trait on a tree*

Description

Computes the Parsimony Score (PS) using a Fitch parsimony procedure for a (typically) binary tree: for each internal node, sets are intersected or unioned; each union corresponds to one inferred change.

Usage

```
calculate_PS(tree, trait)
```

Arguments

`tree`	A phylogenetic tree of class `"phylo"` (assumed bifurcating).
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.

Details

This implementation assumes each internal node has exactly two children. If your tree is not strictly bifurcating, results may be incorrect. See Fitch 1971.

Value

Numeric scalar, the PS value.

References

Fitch, W. M. (1971). Toward defining the course of evolution: minimum change for a specific tree topology.

calculate_pvalues_hypergeom
Compute hypergeometric p-values for subtree (internal)

Description

Compute hypergeometric p-values for subtree (internal)

Usage

```
calculate_pvalues_hypergeom(df_obs)
```

Arguments

df_obs	Data frame returned by `table_info_subtrees_obs()`.

Value

A data.frame with columns node and pvalue.

calculate_pvalues_perm
Computes subtree-level permutation p-values by comparing the observed proportion of `trait1` in each subtree to its null distribution obtained by randomizing tip labels.

Description

Computes subtree-level permutation p-values by comparing the observed proportion of `trait1` in each subtree to its null distribution obtained by randomizing tip labels.

Usage

```
calculate_pvalues_perm(res, df, n_simu)
```

Arguments

res	A list returned by `random_trees_labels()`, containing simulated subtree proportions under the null hypothesis.
df	A data.frame returned by `table_info_subtrees_obs()`, containing observed subtree statistics.
n_simu	Integer. Number of permutations used to build the null distribution.

Value

A list with two elements:

- `pvalues_table`: data.frame with permutation p-values per node.
- `binary_matrix`: binary matrix used for hierarchical correction.

calculate_pvalue_for_AI

Compute permutation p-value for Association Index (AI)

Description

Computes a one-sided permutation p-value for AI as:

$$p = (1 + \sum I(AI_{null} \leq AI_{obs}))/(1 + n_{simu})$$

where smaller AI indicates stronger clustering.

Usage

```
calculate_pvalue_for_AI(ai_obs, res, n_simu)
```

Arguments

`ai_obs`	Observed AI statistic (numeric scalar).
`res`	Output of `random_AI_scores` (must contain `ai_scores`).
`n_simu`	Number of permutations used (must match `length(res$ai_scores)`).

Details

See Wang and al, 2001.

Value

A one-row `data.frame` with columns `statistic` and `pvalue`.

References

Wang and al (2001). Identification of Shared Populations of Human Immunodeficiency Virus Type 1 Infecting Microglia and Tissue Macrophages outside the Central Nervous System.

calculate_pvalue_for_MC

Compute permutation p-values for Monophyletic Clade size (MC)

Description

Computes one-sided permutation p-values for MC (larger MC indicates stronger clustering), separately for `trait0` and `trait1`:

$$p = (1 + \sum I(MC_{null} \geq MC_{obs}))/(1 + n_{simu})$$

.

Usage

```
calculate_pvalue_for_MC(mc_obs, res, n_simu)
```

Arguments

- `mc_obs` Named numeric vector of length 2 with observed MC values for `trait1` and `trait0`.
- `res` Output of `random_MC_scores`.
- `n_simu` Number of permutations used.

Details

See Salemi and al, 2005

Value

A `data.frame` with columns `trait`, `MC_obs`, `pvalue`.

References

Salemi and al (2005). Phylodynamic analysis of human immunodeficiency virus type 1 in distinct brain compartments provides a model for the neuropathogenesis of AIDS.

`calculate_pvalue_for_PS`

Compute permutation p-value for Parsimony Score (PS)

Description

Computes a one-sided permutation p-value for PS as:

$$p = (1 + \sum I(PS_{null} \leq PS_{obs}))/(1 + n_{simu})$$

where smaller PS indicates stronger clustering (more phylogenetic signal).

Usage

```
calculate_pvalue_for_PS(ps_obs, res, n_simu)
```

Arguments

- `ps_obs` Observed PS statistic (numeric scalar).
- `res` Output of `random_PS_scores` (must contain `ps_scores`).
- `n_simu` Number of permutations used (must match `length(res$ps_scores)`).

Details

See Fitch 1971.

Value

A one-row `data.frame` with columns:

statistic Observed PS value.

pvalue Permutation p-value.

References

Fitch, W. M. (1971). Toward defining the course of evolution: minimum change for a specific tree topology.

get_site_candidates *Identify candidate mutations on branches leading to selected clades*

Description

Given one or several internal nodes (typically detected by `mutaphy_test`), this function performs ancestral state reconstruction site-by-site using `ape::ace` (discrete model) and reports genomic positions where a change is detected on the branch leading to each target node. Detection is controlled by a probability threshold comparing parent vs child reconstructed states.

Usage

```
get_site_candidates(nodes, tree, sequences, threshold = 0.3, verbose = FALSE)
```

Arguments

`nodes`	Integer or character vector of internal node identifiers in `tree`.
`tree`	A phylogenetic tree of class `phylo`. Tip labels must match the names of `sequences`.
`sequences`	Named list of aligned nucleotide sequences. Each element is a character vector of nucleotides ("A","C","G","T"), with optional `NA` for missing/ambiguous characters. Names must match `tree$tip.label`.
`threshold`	Numeric. Minimum difference between parent and child posterior probabilities required to call a candidate mutation (default: 0.3).
`verbose`	Logical. If `TRUE`, print diagnostic messages.

Value

A list with element `candidates_by_node`, a named list where each name is a node id and each value is an integer vector of candidate site positions (empty if none detected).

Examples

```
## Not run:
sites <- get_site_candidates(nodes = 360, tree = tree_ids, sequences = seqs, threshold = 0.3)
sites$candidates_by_node[["360"]]

## End(Not run)
```

get_tips_descendants *Get descendant tip indices of a node*

Description

Returns the indices of descendant tips (i.e., values in `1:ape::Ntip(tree)`) for a given internal (or any) node.

Usage

```
get_tips_descendants(tree, node)
```

Arguments

`tree`	A phylogenetic tree of class `"phylo"`.
`node`	Integer node index in the `"phylo"` encoding.

Value

Integer vector of tip indices (subset of `1:ape::Ntip(tree)`).

get_tree_structure *Build the node neighborhood structure used for hierarchical correction (internal)*

Description

Build the node neighborhood structure used for hierarchical correction (internal)

Usage

```
get_tree_structure(tree)
```

Arguments

`tree`	A `phylo` object.

Value

A named list mapping each internal node to its descendants and parent.

mc_test *Permutation test for Monophyletic Clade size (MC)*

Description

Computes the observed MC statistic for two interest states and assesses significance via permutation tests.

Usage

```
mc_test(tree, trait, trait0, trait1, n_simu = 1000)
```

Arguments

tree	A phylogenetic tree of class "phylo".
trait	A vector of tip states (length = ape::Ntip(tree)), ordered as tree$tip.label.
trait0	Character scalar: label for state "0".
trait1	Character scalar: label for state "1".
n_simu	Number of permutations (default: 1000).

Details

See Salemi and al, 2005

Value

A list with:

mc_obs Named numeric vector: observed MC values for trait1 and trait0.

null_distributions List with permuted MC vectors for each state.

pvalues data.frame of permutation p-values for each state.

References

Salemi and al (2005). Phylodynamic analysis of human immunodeficiency virus type 1 in distinct brain compartments provides a model for the neuropathogenesis of AIDS.

See Also

calculate_MC

Examples

```
## Not run:
set.seed(1)
tree <- ape::rcoal(30)
trait <- sample(c(rep("severe", 15), rep("non severe", 15)))

res_mc <- mc_test(
  tree = tree, trait = trait,
  trait0 = "non severe", trait1 = "severe",
```

```
  n_simu = 100
)
res_mc$pvalues

## End(Not run)
```

---

mutaphy_test   *Detect phenotype-associated clades using permutation and hypergeometric tests*

Description

`mutaphy_test()` identifies subtrees enriched for a binary phenotype (encoded in the tip labels of a `phylo` object). It computes both: (i) hypergeometric p-values based on counts within each subtree, and (ii) permutation p-values by randomizing tip labels `n_simu` times. A hierarchical correction procedure is applied to permutation results to account for nested (overlapping) subtrees.

Usage

```
mutaphy_test(
  tree,
  trait1,
  trait0,
  n_simu = 1000,
  alpha = 0.05,
  verbose = FALSE
)
```

Arguments

tree	A phylogenetic tree of class `phylo`. Tip labels must encode the phenotype state (e.g., `trait1` vs `trait0`).
trait1	Character string. Name of the phenotype of interest.
trait0	Character string. Name of the reference phenotype.
n_simu	Integer. Number of permutations used to estimate the null distribution for permutation p-values.
alpha	Numeric. Significance threshold used to define positive subtrees.
verbose	Logical. If `TRUE`, print progress and interpretation messages.

Value

A named list with the following components:

- `tree`: summary statistics at the tree level (min/mean raw and corrected p-values).
- `subtrees`: subtree-level results for both hypergeometric and permutation tests.
- `permutation_matrices`: binary and corrected permutation matrices.
- `positifs`: sets of significant nodes (raw and corrected) and correction parameters.

Examples

```
## Not run:
res <- mutaphy_test(
  tree,
  trait1 = "severe",
  trait0 = "non_severe",
  n_simu = 1000,
  alpha = 0.05,
  verbose = TRUE
)
res$positifs$permutation_nodes_corrected

## End(Not run)
```

---

phylo.stats *Compute phylogenetic association statistics (AI, PS, MC)*

Description

Convenience wrapper to compute three phylogenetic association statistics: Association Index (AI), Parsimony Score (PS), and Monophyletic Clade size (MC), including permutation p-values and null distributions.

Usage

```
phylo.stats(tree, trait, trait0, trait1, n_simu = 1000, alpha = 0.05)
```

Arguments

`tree`	A phylogenetic tree of class "phylo".
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.
`trait0`	Character scalar: label for state "0".
`trait1`	Character scalar: label for state "1".
`n_simu`	Number of permutations used for PS and MC (default: 1000).
`alpha`	Significance level (currently not used; kept for convenience).

Details

`AI` is computed via `ai_test` (which has its own default `n_simu = 1000`). In the current implementation, `phylo.stats()` does not pass `n_simu` to `ai_test()`.

Value

A nested list with components `AI`, `PS`, and `MC`. Each contains the statistic(s), p-value(s), and corresponding null distribution(s).

References

PS: Fitch, W. M. (1971). Toward defining the course of evolution: minimum change for a specific tree topology. AI: Wang and al (2001). Identification of Shared Populations of Human Immunodeficiency Virus Type 1 Infecting Microglia and Tissue Macrophages outside the Central Nervous System. MC: Salemi and al (2005). Phylodynamic analysis of human immunodeficiency virus type 1 in distinct brain compartments provides a model for the neuropathogenesis of AIDS.

See Also

ai_test, ps_test, mc_test

Examples

```
## Not run:
set.seed(1)
tree <- ape::rcoal(30)
trait <- sample(c(rep("severe", 15), rep("non severe", 15)))

phylo_stat <- phylo.stats(
  tree = tree, trait = trait,
  trait0 = "non severe", trait1 = "severe",
  n_simu = 100
)
phylo_stat$AI$pvalue
phylo_stat$PS$pvalue
phylo_stat$MC$pvalues

## End(Not run)
```

---

ps_test — *Permutation test for Parsimony Score (PS)*

Description

Computes the observed Parsimony Score (PS) and assesses significance via a permutation test.

Usage

```
ps_test(tree, trait, n_simu = 1000)
```

Arguments

tree	A phylogenetic tree of class "phylo".
trait	A vector of tip states (length = ape::Ntip(tree)), ordered as tree$tip.label.
n_simu	Number of permutations (default: 1000).

Details

PS is computed using a Fitch parsimony-style recursion on a binary tree. Lower PS indicates fewer changes and thus stronger clustering of identical states. See Fitch 1971.

Value

A list with:

ps_obs Observed PS value.

pvalue Permutation p-value (numeric scalar).

null_distribution Numeric vector of permuted PS values.

References

Fitch, W. M. (1971). Toward defining the course of evolution: minimum change for a specific tree topology.

See Also

calculate_PS

Examples

```
## Not run:
set.seed(1)
tree <- ape::rcoal(30)
trait <- sample(c(rep("severe", 15), rep("non severe", 15)))

res_ps <- ps_test(tree = tree, trait = trait, n_simu = 100)
res_ps$pvalue

## End(Not run)
```

---

random_AI_scores — *Generate null distribution for Association Index (AI) by permutation*

Description

Computes a null distribution of the Association Index (AI) by permuting the trait labels on the tips `n_simu` times and recomputing AI each time.

Usage

```
random_AI_scores(tree, trait, n_simu = 1000)
```

Arguments

`tree`	A phylogenetic tree of class "phylo".
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.
`n_simu`	Number of permutations (default: 1000).

Details

See Wang and al, 2001.

Value

A list with:

ai_scores Numeric vector of length `n_simu` containing permuted AI values.

References

Wang and al (2001). Identification of Shared Populations of Human Immunodeficiency Virus Type 1 Infecting Microglia and Tissue Macrophages outside the Central Nervous System.

See Also

`calculate_AI`, `ai_test`

`random_MC_scores` *Generate null distributions for Monophyletic Clade size (MC) by permutation*

Description

Computes null distributions for the Monophyletic Clade (MC) statistic for two states (`trait0` and `trait1`) by permuting tip labels and recomputing MC.

Usage

```
random_MC_scores(tree, trait, trait0, trait1, n_simu = 1000)
```

Arguments

`tree`	A phylogenetic tree of class "phylo".
`trait`	A vector of tip states (length = `ape::Ntip(tree)`), ordered as `tree$tip.label`.
`trait0`	Character scalar: label for state "0" (or the reference state).
`trait1`	Character scalar: label for state "1" (or the interest state).
`n_simu`	Number of permutations (default: 1000).

Details

See Salemi and al, 2005

Value

A list with:

mc_trait0 Numeric vector of length `n_simu`: permuted MC for `trait0`.

mc_trait1 Numeric vector of length `n_simu`: permuted MC for `trait1`.

References

Salemi and al (2005). Phylodynamic analysis of human immunodeficiency virus type 1 in distinct brain compartments provides a model for the neuropathogenesis of AIDS.

See Also

`calculate_MC`, `mc_test`

random_PS_scores *Generate null distribution for Parsimony Score (PS) by permutation*

Description

Computes a null distribution of the Parsimony Score (PS) by permuting the trait labels on the tips n_simu times and recomputing PS each time.

Usage

```
random_PS_scores(tree, trait, n_simu)
```

Arguments

tree	A phylogenetic tree of class "phylo" (typically rooted and binary).
trait	A vector of tip states (length = ape::Ntip(tree)), ordered as tree$tip.label.
n_simu	Number of permutations used to build the null distribution.

Details

The permutation test breaks any phylogenetic association by randomly shuffling trait labels across tips. See Fitch 1971.

Value

A list with one element:

ps_scores Numeric vector of length n_simu containing permuted PS values.

References

Fitch, W. M. (1971). Toward defining the course of evolution: minimum change for a specific tree topology.

See Also

calculate_PS, ps_test

random_trees_labels *Permute tip labels to generate null subtree statistics (internal)*

Description

Permute tip labels to generate null subtree statistics (internal)

Usage

```
random_trees_labels(tree, n_simu, trait1)
```

Arguments

tree	A phylo object.
n_simu	Integer. Number of permutations.
trait1	Phenotype of interest.

Value

A list of matrices containing simulated subtree proportions and counts.

---

table_info_subtrees_obs

Compute subtree statistics for an observed phenotype (internal)

Description

Compute subtree statistics for an observed phenotype (internal)

Usage

```
table_info_subtrees_obs(tree, trait1)
```

Arguments

tree	A phylo with phenotype encoded in tip labels.
trait1	Phenotype of interest.

Value

A data.frame with counts and proportions of trait1 for each internal node.